

EXECUTIVE BRIEF • JULY 2026 • 7 MIN

QA catches the mistake. The process still loses eight days.

I built an agent to move the same check earlier – before the queues, handoffs, and re-review begin. It reads the record, checks the handbook, and shows the evidence behind every answer. The AI gives an answer. The system makes it prove it. People handle the hard calls.

The scale about a third of a million verifications in six months • **The cost** roughly eight days from QA return to approval • **The status** built, tested, and entering reviewer feedback

THE COST OF CATCHING IT LATE

The error is caught. The delay comes after.

QA reviewers react quickly. Rework usually begins within an hour of a return. But the record then moves through queues, handoffs, corrections, and another review. By the time it returns to approval, roughly eight days have passed.

The agent does not make that loop faster. It is built to stop the loop from starting. In the first half of 2026, about 2% of finished verifications returned with a coded error – a floor, not a census – and about 86% of coded return reasons fell into four procedural categories: routing, discrepancy classification, incomplete fields, and note handling. Those are rule-shaped mistakes a machine can check when the draft is created.

WHAT I WON'T CLAIM

Those are **elapsed** times, not effort – nothing in this data measures the minutes a reviewer actually spends on a case, so I make no labor-savings claim until activity logs exist to prove one. What is measured is the calendar cost of the return loop; the harness attacks that cost by construction, and the same script that produced this baseline will count the detours it deletes.

WHY NOW

Every wave of automation sends more work to the checker

QA already touches 95% of searches. As fulfillment becomes more automated, the final check does not disappear – it becomes a larger share of the work people still do.

That makes QA the next constraint, and the place where this agent matters.

THE FIX

So I moved the check earlier

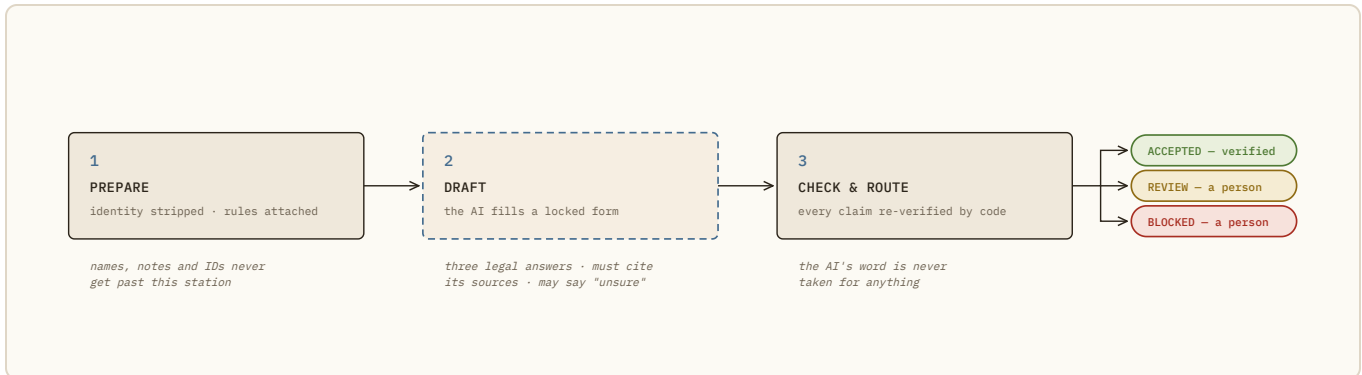
The agent reviews each completed education and employment verification before it enters the return loop. It has three possible answers: this is correct, this needs a fix, or a person needs to look.

It can inspect the timeline, check the evidence, and consult the handbook. Every answer must show its work. If the evidence is missing, the rules do not support the answer, or the case genuinely requires judgment, it goes to a reviewer.

It learned the job from the operation's own record. The QA handbook was codified into machine-enforceable rules – the agent must cite them on every draft, and a citation to a rule that doesn't exist is an automatic block, not a deduction. A year of historical QA decisions

became its benchmark and its first precedents: every score in this brief is the agent measured against what human reviewers actually did on real cases.

Three stations, one way through:



The case is prepared, drafted, checked, and routed. Even an accepted repair is applied by a reviewer; the system prepares the answer and proves it, but does not push the button.

UNDER THE HOOD

What is actually under the hood?

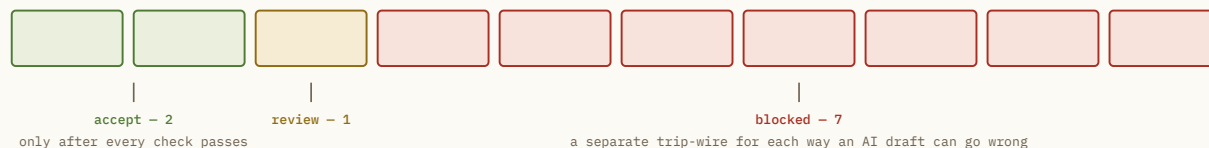
The agent is a small language model surrounded by rules, evidence checks, and hard stops. The model reasons across a messy case; the harness decides whether its answer may survive.

Before the model sees a case, names, free-text notes, and identifiers are stripped, hashed, or reduced to flags. The model then returns one of three answers on a locked form. Every citation, evidence claim, confidence score, and actionable code is rechecked before the case can leave.

It drafts. The harness checks. Anything uncertain stops with a person.

THE SAFETY ARCHITECTURE

More ways to say no than yes



A case leaves the system stamped with one of exactly ten outcomes. Seven of the ten are refusals – one each for leaked personal data, malformed output, invented citations, unsupported repair claims, invented code meanings, low confidence, and missing evidence. A blocked draft is stripped of anything executable before it's re-routed to a person.

The model can suggest an answer. It cannot quietly approve one. The design rule underneath fits in one line: **when anything doesn't add up, the case goes to a person – and no other path exists.** Every failure, ambiguity, or malfunction inside the system collapses to its safest state – a human review – the way an elevator brake clamps when power is cut. Not because a policy says so, but because the software has no other exit. The AI can be wrong; it cannot be *silently* wrong.

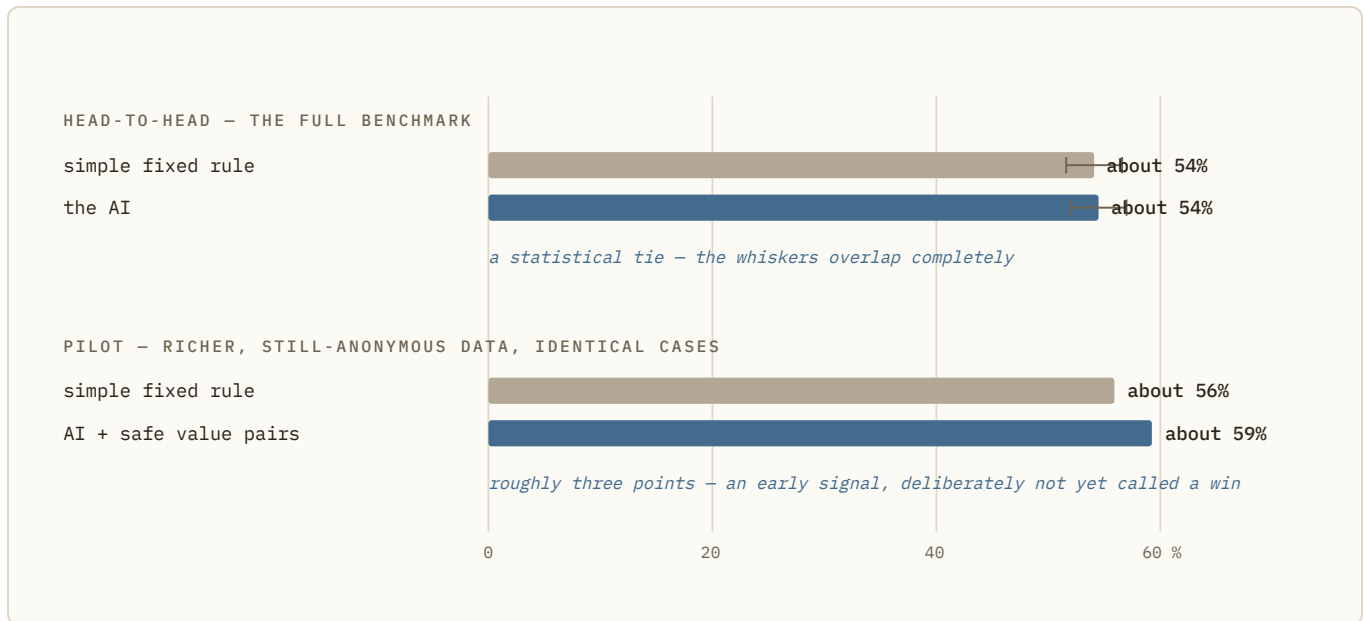
TAKEAWAY

Six independent checks ran on every benchmark case – and **zero unsupported commitments escaped them.** The promise is not “trust the AI.” It is “make every answer prove itself.”

THE SCOREBOARD

Where it stands today

The head-to-head that matters: on just over two thousand historical cases, did the AI predict the reviewer's action better than a deliberately simple fixed rule?



On the full benchmark, the AI and the fixed rule tied at about 54% – the whiskers show sampling noise, and the bars overlap completely. With richer but still anonymous evidence, the comparison moved to about 59% versus 56%: an early signal, still within the uncertainty of this dataset. "Safe value pairs" means job titles, degrees and dates – never names, contacts or identifiers; exact figures live in the deep-dive appendix.

On a side-by-side test, the local and hosted versions gave the same answer on every case. That makes local experimentation useful and inexpensive, with the final candidate checked again in the hosted lane.

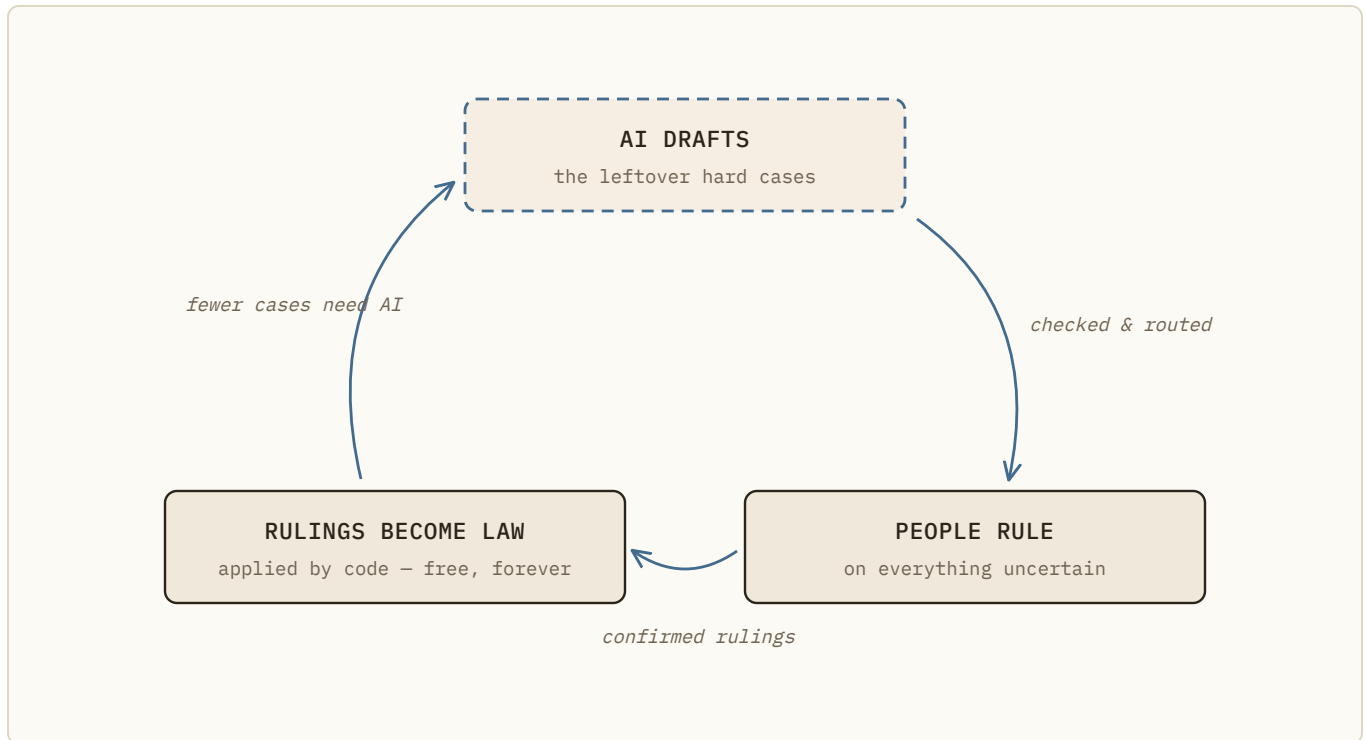
TAKEAWAY

The full benchmark was a tie. Richer safe evidence created a small early lead; better evidence mattered more than another clever prompt.

The rules, audit trail, benchmark, and sealed blind test now exist. That is enough to move from demonstration to reviewer feedback. The next evidence comes from the first cases reviewed by the people who know the work.

THE LEARNING LOOP

It learns as it goes



The learning loop. Reviewers settle the hard calls, rulings become precedent, and repeat cases move into code – the latest benchmark queued precedent candidates from roughly 3% of its cases.

Today, a person still applies every correction. Over time, each confirmed ruling moves another repeat question into code, leaving reviewers with the cases that are genuinely new.

At the finish line, the agent checks every case, people review by risk, and senior reviewers own disputes and new precedent.

The next step is simple: review the first cases and start turning confirmed rulings into reusable answers.

TO REMEMBER

If you keep three things

-
- 01 **The AI gives an answer. The system makes it prove it. People handle the hard calls.**
 - 02 **As more fulfillment automates, QA becomes the human constraint this agent is built to relieve.**
 - 03 **Every confirmed ruling becomes precedent, so repeat questions move from AI into code.**
-

Internal briefing · 2026-07-03. Companion reading: the conceptual tour (how the architecture works, room by room) and the implementation deep dive (module names, schemas, and the source of every number cited here).

All figures trace to repository records as of July 3, 2026; the operational baseline traces to analysis/qa_ops_metrics_20260707/. Operating-model figures – the automation curve, review-touch shares, review-tax and sampling envelopes, and the design targets for the finish line – trace to the Verifications Operating Model Final Report (2026-07-02) and its capacity model; modeled targets are labeled as modeled, measurement windows are labeled, and figures from different windows are never cross-divided. The learning-loop shares shown in diagrams are illustrations, not measurements. No applicant data appears in this document.

SNH CAPITAL PARTNERS · A Wright Original · Built by SNH Strategic Initiatives

CONCEPTS · JULY 2026 · 12 MIN READ

A machine that knows when it doesn't know

Follow one verification through five rooms. Watch its identity disappear, its evidence take shape, the model draft an answer, and the system decide whether that answer deserves to leave the building.

QA repair harness team · companion to the [implementation deep dive](#)

Every automated decision system reveals itself when it is unsure. Some guess. Some hide uncertainty behind polished language. This one stops, records why it stopped, and hands the case to a person.

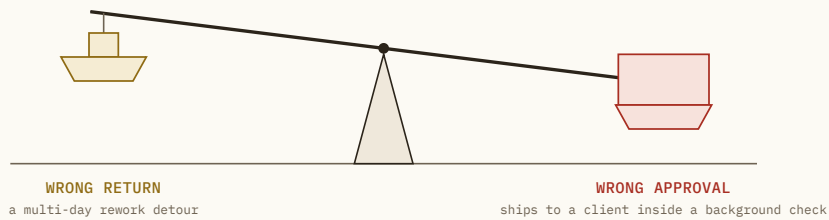
THE TRAVELING CASE

The provided employment dates and the verified dates differ. The record may need a return — or the difference may be harmless. And one piece of evidence is missing.

THE PROBLEM

Two ways to be wrong, one of them expensive

A reviewer can be wrong in two directions. Send a correct record back and it takes a measured detour — roughly eight days before approval returns. Approve an incorrect record and the error ships to a client inside a background check. In the first half of 2026, about 2% of finished verifications returned with a coded QA error — a floor, not a census — and most were procedural and rule-shaped.



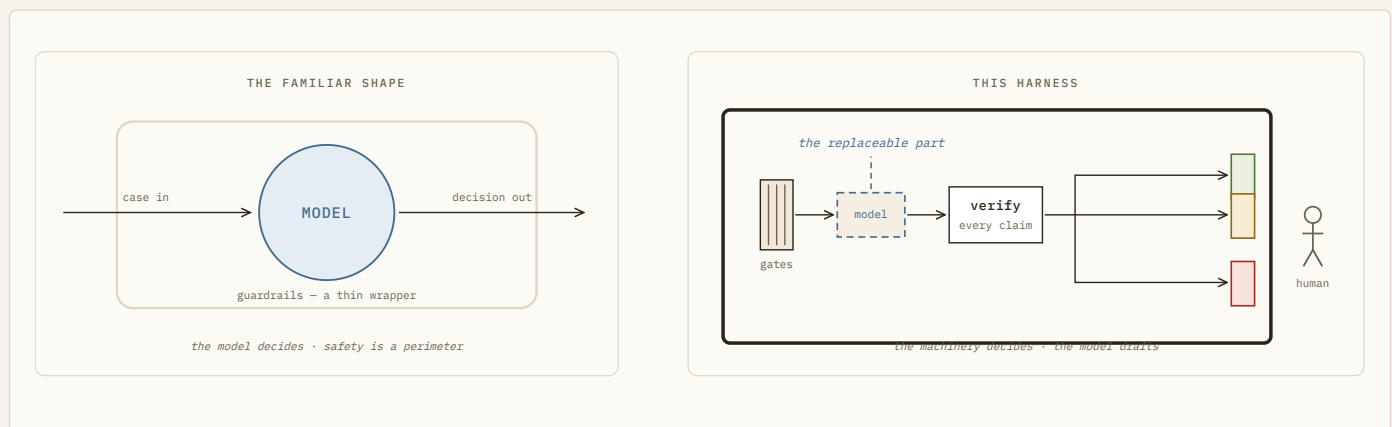
The failure asymmetry. The entire harness is engineered so that when it errs, it errs on the annoying side – never the expensive one. The house rule: like an elevator brake that clamps when power is lost, every failure inside the system collapses to its safest state – which here means "a human looks at it."

A language model raises the stakes because its wrong answer can be confident, fluent, and perfectly formatted. The architecture therefore cannot be "ask the model and trust it when it sounds sure."

THE IDEA

Put the model in the smallest room

Most systems that put an LLM to work share a familiar shape: the model sits in the middle, the decision flows through it, and safety is a perimeter – filters and guardrails wrapped around a trusted core. This harness inverts that. The decision-maker is a stack of plain, predictable software – gates, precedent lookups, a verifier that re-checks everything – and the model is a small, sealed, swappable box off to one side that produces *drafts*.



The model supplies a draft. Everything around it decides whether the draft survives. In the familiar shape, intelligence and authority live in the same place; here they are kept apart.

The codebase gives the game away in its vocabulary: the model component is called the *reasoner*, never the *decider*. And the very first sentence the model reads, on every single case, is this:

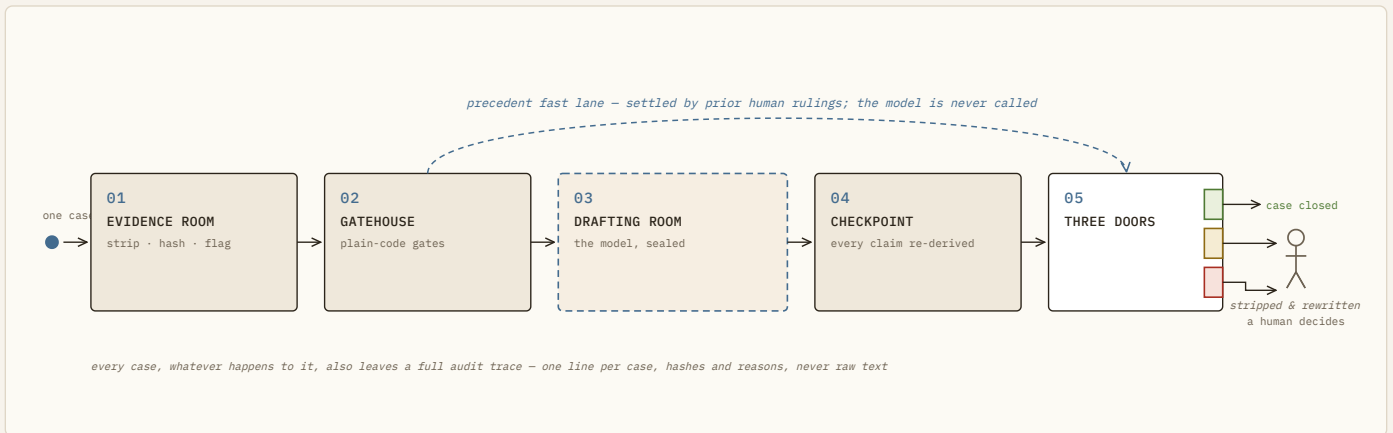
"You are the replaceable reasoning component in a rule-governed QA repair harness."

– the production system prompt, opening line

THE TOUR

Follow one case through the building

Five rooms, one case, and a fast lane for questions people have already settled.



The floor plan. Note where the model sits: room 3 of 5, dashed walls, with machinery before it deciding whether it's even needed and machinery after it deciding whether to believe it.



ROOM 1

Keep the facts. Lose the identity.

THE CASE · ROOM 1

Loses its identity, keeps its shape. The name is gone, the note is now “note present,” the id is a hash – what remains: two date ranges that differ, and one evidence flag missing.

Room 1 turns the warehouse row into a *fact packet*: the case keeps its logical shape and loses its identity.

AS THE WAREHOUSE HOLDS IT		AS EVERYTHING DOWNSTREAM SEES IT
applicant name	████████	- never leaves the warehouse
reviewer note	██████████████	note_present: true
search id	██████	a1f3...9c04 (salted hash)
provided vs verified	██████████	family: date_range · differs: yes
fields that disagree	████████	count: 2

Free text becomes yes/no flags. Identities become hashes. Values become families. The model will later reason about the **shape of a discrepancy**, not the person behind it.

The same privacy screen runs before every hosted request and after every model response.
The hosted connection refuses any payload richer than the stripped tier. Privacy is checked by the system, not promised by the team.

□ □ **■** □ □

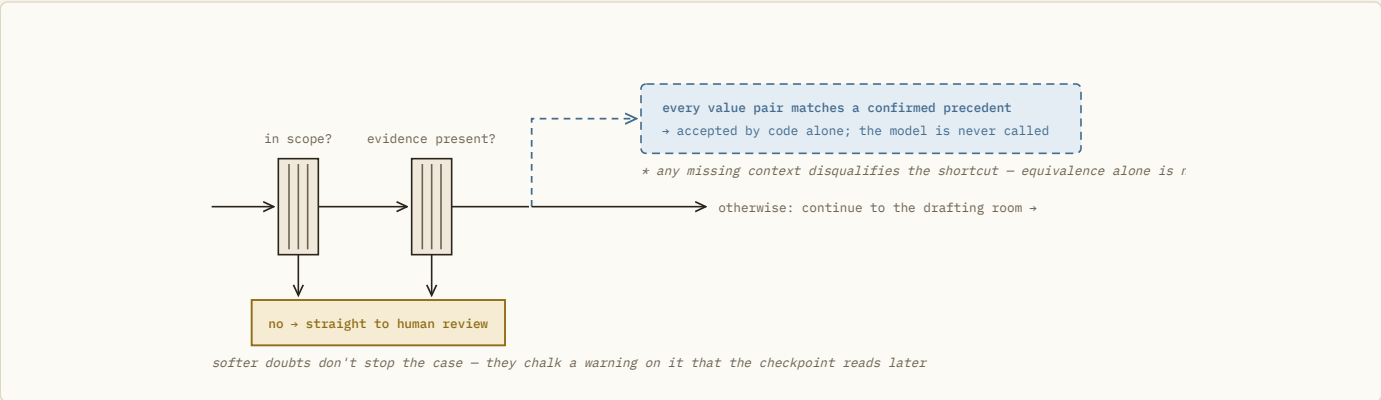
ROOM 2

Let the cheap questions go first.

THE CASE · ROOM 2

Gains its rule scope — and a chalk mark: “evidence gap.” No confirmed precedent matches these exact values and the context is incomplete, so the fast lane is closed. The case walks on.

Plain code asks the cheap questions first: is the case in scope, and does the required evidence exist? A hard failure goes directly to a person. A complete match to a human-confirmed precedent takes the fast lane through code. Missing context closes the shortcut and sends the case onward.



The gatehouse. Hard questions end the trip; soft doubts travel with the case as chalk marks. The fast lane is the precedent ledger – prior human rulings applied by code, the same way every time.

□ □ **■** □ □

ROOM 3

Give the model a form, not a microphone.

THE CASE · ROOM 3

Gains a draft. The model proposes “needs a fix,” citing the date rule and the mismatch it can see.

Only now does the model enter the story, and it works the way a contractor works in a secure facility: in a sealed room, with only the documents handed through the slot, filling out a form.



The drafting room. Everything the model knows arrived through the slot; everything it says leaves as one machine-checkable form.

The model must return one of three answers: fix it, accept it, or send it to a person. Decoding enforces the form, and every claim must cite the allowed rules and evidence. A better model can later occupy the same slot without changing the surrounding controls.

The model can request one of seven read-only lookups: timeline, evidence, code meanings, safe comparisons, rules, and product context. Every answer returns with a receipt the verifier can inspect. Repeat a question, request something unavailable, or exhaust the budget, and the case goes to a person. Whatever happens, the model still leaves the room with the same locked form.



ROOM 4

Never grade your own homework.

THE CASE · ROOM 4

Gains six stamps – five green, one red. The proposed fix leans on evidence the packet doesn't contain: the chalk mark from room 2 just became a hard refusal.

The draft now meets the verifier, and the verifier's stance is the whole system in miniature: **it doesn't ask whether the draft sounds right; it re-derives every checkable claim from the same inputs.** Six stamps, all of which must land:

✓ Citations exist. Every id the model cited exists on the allowed list it was handed. An invented citation isn't a deduction – it's an automatic block.	✓ Claims match evidence. A "fix it" or "it's fine" must rest on at least one rule <i>and</i> one piece of evidence – and the cited rule must actually govern the outcome claimed.	✓ Confidence clears the floor. Below the 65% confidence threshold, the model may not commit either way. It can only recommend a human.
✓ Nothing sensitive appears. The same privacy screen from room 1 runs over the model's own output – in case the model hallucinated something that looks like personal data.	✓ Repair claims have proof. "This was already corrected" requires field-level before/after proof upstream. Without it, the claim cannot stand.	✓ Code meanings are known. Status codes with no authoritative mapping may be echoed, never interpreted. The model doesn't get to guess what code 7 means.

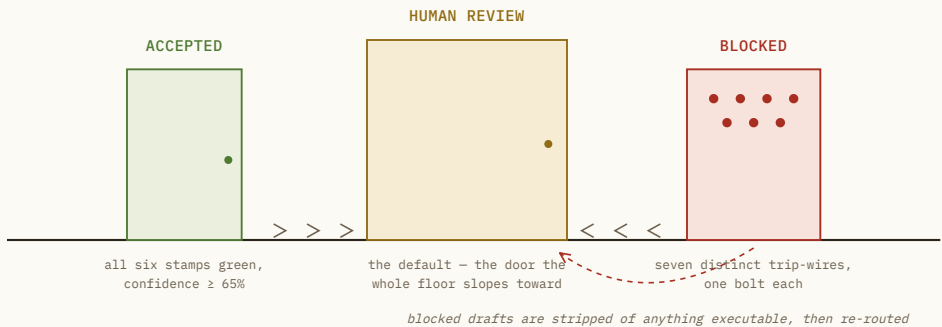


ROOM 5

Three doors, with the floor tilted toward a person.

THE CASE • ROOM 5

Gains its route: rewritten to human review, proposed codes stripped, the reason attached. A person decides, with the whole trail in hand.



The exits. Ten labeled outcomes hide behind these three doors – and seven of the ten are variations of the red one. Only the green door clears the harness without another judgment call. Even then, a person applies the correction.

The green door opens only when every check passes. Human review is the default. A blocked draft is rewritten before routing: its verdict becomes human review, its actionable codes are removed, and the reason travels with the case. No route produces a silent approval.

TRY IT

You be the router

The routing precedence is easier to feel than to memorize. Below is a draft decision fresh out

of the drafting room. Flip its properties and watch which door the case leaves through.

ROUTE A DRAFT – LIVE

- ☒ Output free of anything that looks like personal data
- ☒ Every citation comes from the allowed list
- ☐ Backed by at least one rule and one piece of evidence
- ☒ Confidence at or above the 65% floor

The model's own verdict: needs a fix

● BLOCKED – no material support

A committed verdict needs at least one rule and one piece of evidence behind it. Fluent reasoning with nothing underneath doesn't clear the checkpoint.

SAFETY CITES GROUND CONF REPAIR CODES

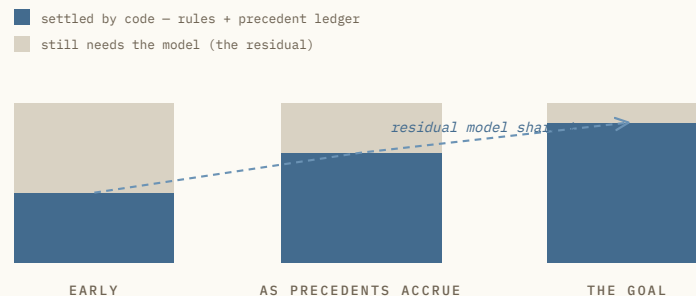
DESTINATION: the draft is rewritten on the spot – verdict overwritten to human review, actionable codes stripped – and the case lands in a reviewer's queue carrying nothing executable.

The traveling case begins blocked because its proposed fix lacks evidence. Repair the grounding and the door turns green. The checks run in severity order – privacy, citations, evidence, then confidence – and **no combination of switches produces a silent approval**.

THE MEMORY

Every confirmed answer is one less question for the model

When a human confirms a ruling, it becomes precedent. The next matching case takes the fast lane through code, and the model's jurisdiction shrinks by one case pattern. Instead of trying to make the model trusted with more, the system moves every hard-won certainty into code – free, immediate, and auditable.

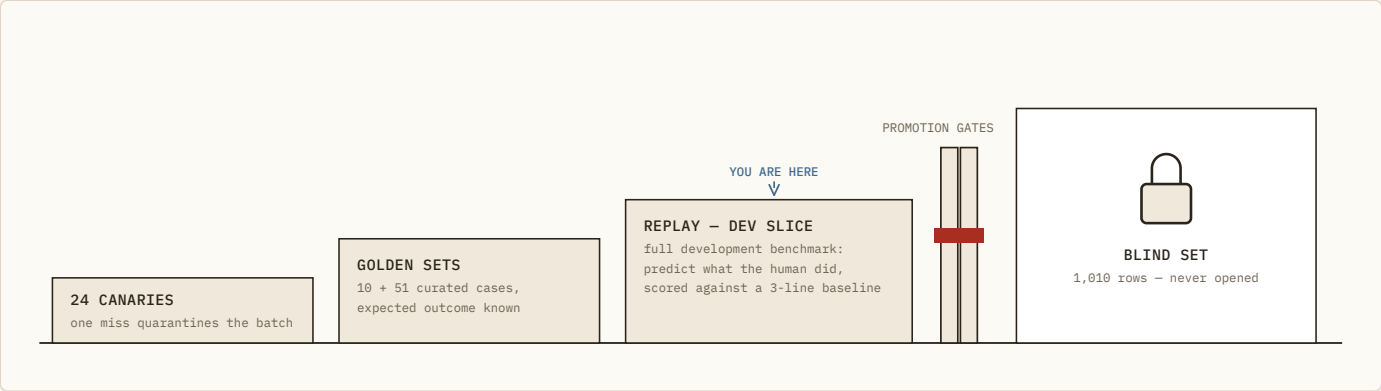


An illustration, not a measurement. The ledger begins with a small handbook-seeded set; every confirmed ruling moves matching cases from the model share to the settled-by-code share. The latest benchmark surfaced precedent candidates in roughly 3% of cases.

THE PROOF

The final test stays sealed

The model does not get to practice on the final exam. The evaluation is built with the same suspicion as the pipeline:



The trust ladder. Each step is cheaper to fail than the next. The final room stays locked until a candidate clears every gate – and none has.

The staircase starts with **canaries** – synthetic cases with known answers that run before every batch; a single miss quarantines the batch. It ends at a **blind set** the team has deliberately never scored against, reserved for the day a candidate earns it. In between sits the replay benchmark, whose honesty features are worth naming: the model predicts from exactly the information the original human reviewer had (everything they wrote afterward is withheld), the labels are treated as "what the human did," not "the truth," and every score is compared against a deliberately embarrassing baseline – a three-line fixed rule.

≈ 54% THE THREE-LINE FIXED RULE	≈ 54% THE MODEL	≈ a tie THE FINDING – AND THE SYSTEM SAID SO ITSELF
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That tie is the most credible thing about the project. A lesser evaluation setup would have produced a happier number – and early, smaller runs did, until scale washed the mirage out. The harness's own instruments caught it: gates held, every unsupported commitment was blocked rather than shipped, and the disagreements between model and rule were harvested as new precedent candidates for the ledger. **A benchmark you can't fail is marketing. This one can say "not yet," and it has.**

IN ONE SENTENCE

The shape of the idea

Strip identity at the door. Settle cheap questions in code. Let the model investigate and draft on a locked form. Rebuild every claim. Route uncertainty to a person. Turn confirmed rulings into precedent.

Internal engineering note · 2026-07-03 · conceptual companion to the [implementation deep dive](#), which carries the module names, schemas, and sourced numbers behind every claim here.

All case content in the illustrations is synthetic or redacted-by-construction. No applicant data appears in this page – the same guarantee the harness enforces on itself.

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ENGINEERING · JULY 2026 · 13 MIN READ

The AI investigates. The harness makes the result trustworthy.

Give the agent a messy verification and it can decide what evidence to inspect, connect the facts to the handbook, and explain what should happen next. Every lookup leaves a receipt. Every claim is checked independently. The result is AI reasoning the operation can actually use.

QA repair harness team · education & employment verifications

Rules handle the known. Reviewers navigate what they have not seen. This agent is built for the space between.

Give it an employment verification with conflicting dates, incomplete evidence, and several possible handbook rules. It does not simply classify the row. **It investigates.**

First it asks for the timeline. Then it checks which evidence exists. It retrieves the rule governing date discrepancies. It notices that the missing primary-source evidence prevents a confident correction. Its final answer is not a guess: send the case to a person, with the rule, the evidence, and the missing proof attached.

AI MOVE	WHAT IT LEARNS
Show timeline	provided and verified dates differ
Check evidence	primary-source proof is missing
Retrieve rule	a committed correction requires supporting evidence
File decision	human review, with the missing proof identified

This is not autocomplete. It is an evidence-seeking sequence in which each answer determines the next question. (Synthetic, like every worked example on this page.)

Give it the missing source and the same investigation continues: compare the dates, apply the discrepancy rule, file a supported return, and attach the evidence.

The AI is not designed to stop on every difficult case. It is designed to go exactly as far as the evidence supports.

THE ENABLER

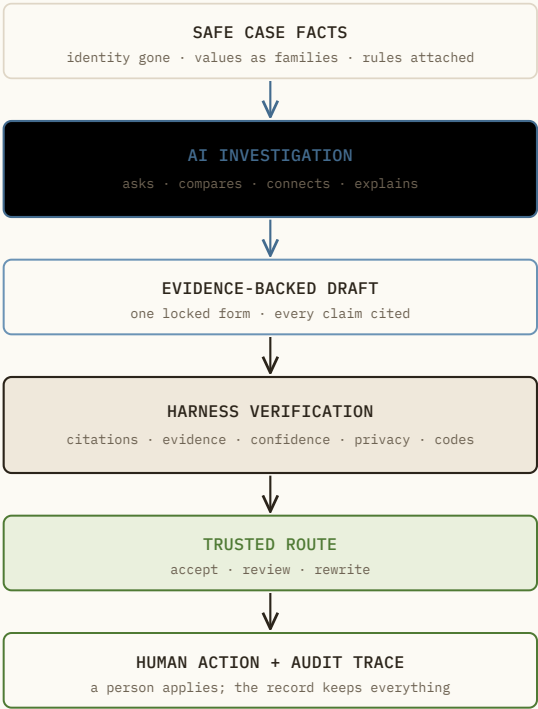
The harness gives the intelligence somewhere safe to work

The model reasons inside a deliberately prepared world: safe facts, relevant rules, read-only tools, and a clear decision contract. The harness prepares that world, checks the resulting argument, and decides what may leave.

That division of labor is the product. **The AI provides the intelligence. The harness gives that intelligence a trustworthy way to operate.**

PEOPLE OWN

Genuine judgment calls, new precedents, and the final application of any correction.



*blue = the AI reasoning path · ink/green = the harness trust path
two systems working together – not one system policing a bad model*

The division of labor, drawn: the intelligence in the middle, the guarantees around it.

THE WORLD, PRECISELY

The model can reason freely inside a world the harness defines

- 1 • **What it sees.** Identity and free text are removed before the model enters.
- 2 • **What it can ask.** Seven read-only tools; no writes, updates, or outside actions.
- 3 • **Which rules exist.** The harness supplies the only rules the model may cite.
- 4 • **What it may return.** Three legal answers on one fixed form.
- 5 • **What counts as proof.** Every committed answer needs a real rule and real evidence.
- 6 • **What happens next.** The harness – not the model – chooses the final route.

Everything before the model prepares its world. Everything after it verifies what survives.

► THE EIGHT STAGES, MODULE BY MODULE

Control one, concretely – what reaches the model:

IN THE WAREHOUSE	WHAT TRAVELS
Applicant name	removed
Reviewer note	<code>note_present: true</code>
Search ID	salted hash
Provided/verified dates	<code>date_range</code> – differs
Allowed evidence	<code>EV-SEG-0004</code>
Allowed rules	<code>R-QA-EMP-DISC-001</code>

THE CONTRACT

Every answer arrives with its evidence attached

The decision is a signed form, not a chat reply. The AI must choose one of three answers – fix it, accept it, or send it to a person – and attach the rule, evidence, requested correction, confidence, and review requirement. Decoding guarantees a valid form; the verifier determines whether its reasoning earns a committed route.

```

{
  "schema_version": "qa_repair_decision.v1",
  "decision": "needs_fix",
  "issue_type": "provided_verified_mismatch",
  "rule_ids": ["R-QA-EMP-DISC-001"],
  "citations": ["R-QA-EMP-DISC-001", "EV-SEG-0004"],
  "what_needs_fixing": "Verified employment dates disagree with the provided range.",
  "how_to_fix_it": {
    "field": "employment_dates",
    "current_value_family": "date_range",
    "correct_value_family": "date_range",
    "current_value": null,
    "correct_value": null
  },
  "correct_output": {
    "result": "discrepancy",
    "qa_action": "qa_return",
    "qa_comment": "Return to verifier: dates require re-verification.",
    "result_id": null,
    "comment_id": null,
    "error_level": null
  },
  "why": "R-QA-EMP-DISC-001 requires a return when verified dates materially differ",
  "confidence": 0.82,
  "review_required": false,
  "warnings": []
}

```

VERIFIED · 6/6 CHECKS

A synthetic example. The nulled `current_value`/`correct_value` are the export-safe tier at work – value families travel, values don't. The nulled catalog codes are the model declining to guess IDs; had it guessed, the verifier would have checked them against the approved result and comment catalogs and blocked any miss.

► THE FULL CONTRACT: DEFAULTS AND DECODING

THE TOOLS

The model can ask questions. It still cannot act.

The one-shot reasoner receives the safe packet at once. The tool-using reasoner builds the same view through read-only lookups, one question at a time. Both return the same locked form and face the same verifier; only the one-shot path has completed the full benchmark.

The menu is seven read-only lookups: **timeline · evidence flags · result-code meaning · comment-code meaning · safe value comparison · rule lookup · product context.**

► THE SEVEN LOOKUPS, PRECISELY

The exciting part is not that the tools are restricted. It is that the model can conduct a small investigation: it chooses what to inspect next, reads the answer, updates its view, and continues until it can file a decision or identify what remains unknown.

Every lookup leaves a receipt. The verifier holds the model to what those receipts actually contain.

Repeated questions, invalid tools, unsafe output, or an exhausted budget all end the same way: stop and send the case to a person.

BENCHMARK BOUNDARY

Every score on this page belongs to the one-shot reasoner. The tool-using loop inherits the controls but has not yet faced the full benchmark. When it has, this page will say so, with numbers.

AFTER THE MODEL

The AI builds the argument. The harness checks every step.

The AI produces the argument. The verifier rebuilds each check from the same case facts, rules, and receipts. Six checks run in a fixed order – data safety first:

<code>data_safety_ok</code> A recursive sensitive-pattern scan (emails, phones, SSNs, URLs, filenames) over the entire decision object.	<code>citation_membership_ok</code> Every citation must be a member of the selected rules $\cup$ evidence ids. Invented citations die here, not in a postmortem.	<code>material_support</code> A committed decision must cite at least one rule <i>and</i> one piece of evidence – and the cited rule must actually govern the outcome claimed.
<code>confidence_ok</code> <code>needs_fix</code> or <code>correct_as_is</code> below 0.65 confidence is refused: <code>confidence_too_low_for_final_decision</code> .	<code>direct_correction_ok</code> "It was directly corrected" requires field-level old/new proof, which the normalizer currently never grants. The claim stays candidate-only.	<code>unknown_mapping_ok</code> Observed <code>qastatus</code> / <code>errorcategoryid</code> codes may be echoed as values under caveat – never interpreted as meaning.

This is where trust is created – and the next section shows it catching a real mistake.

THE FAILURE

Clever does not mean correct

On one case, the model gave a confident answer, cited one real rule, invented another, and attached executable codes. **The harness caught it.**

MODEL SAID	HARNESS DID
<code>needs_fix</code>	rewritten to <code>needs_human_review</code>
included an invented rule	named and blocked the invented citation
attached executable codes	stripped every actionable field
confidence <code>0.81</code>	preserved as evidence, not authority

The invented citation failed the first verification check. The finalizer changed the verdict to human review, removed every actionable code, and recorded the reason. **The AI's reasoning remained visible; the unsupported action did not.**

► INSPECT THE EXACT REWRITE

Every case leaves through one of ten named routes – two accepted, one review, seven blocked. Nothing writes to the warehouse; an accepted repair leaves as an export a human reviewer applies.

► INSPECT ALL TEN TERMINAL ROUTES

THE BLINDFOLD

What the model never sees

Assume every prompt will eventually appear somewhere unintended. The hosted path therefore receives no names, contacts, identifiers, free text, or raw values. The same scanner runs before send and after return; any hit stops the case.

Two payload tiers enforce the boundary: an export-safe tier for hosted models and a richer local-only tier for controlled evaluation. A missing tier is treated as local-only and refused.

► THE PAYLOAD BOUNDARY, PRECISELY

FIELD NOTE – WHEN THE GUARD DOG BITES YOU

The sensitive-text scanner once blocked 12 of 320 replay rows before any model was called. The phone-number regex, written with naive digit boundaries, was matching ten-digit runs *inside the harness's own 20-character hex hashes* – the very hashes the normalizer substitutes for raw identifiers. The guard was eating the disguise it had issued.

The fix was hex-aware boundaries. The lesson was bigger: the safety guards need their own tests, because a paranoid guard also defaults to "send it to a human" – silently, consistently, and at a 4% tax you won't notice unless you're counting.

THE PAYOFF

The model is the replaceable part

The harness does not depend on one model being permanently best. A better model can take the same safe facts, tools, rules, and decision contract – and face the same independent checks.

The intelligence can improve without rebuilding trust from scratch.

THE ENGINE

The boring path is the best path

The AI handles unfamiliar questions. Once people settle one, it graduates into precedent, and the next matching case moves through code.

The precedent ledger turns confirmed rulings into code. A complete match with complete context skips the model entirely; missing context closes the shortcut. The ledger starts with 24 handbook-seeded pair rulings – the seeds establish the mechanism, and reviewer-confirmed rulings are what compound it. The eval splits every decision – settled by precedent, settled by the gates, or needing the model – and tracks one number: how often the model was needed at all.

Every confirmed precedent makes the next matching case faster, cheaper, and easier to audit.

THE BASELINE

The loop being shortened, measured first

Before claiming the harness shortens QA, I measured the existing loop – one reproducible script over hundreds of thousands of finalized education and employment verifications from the first half of 2026. Rework usually begins in under an hour; approval returns roughly eight days later.

WHAT THE CLOCK MEASURES	TYPICAL ELAPSED TIME
QA return to first rework	under an hour
QA return to later approval	about eight days
First return to first approval	about eleven days
Search posted to result sent	about four days

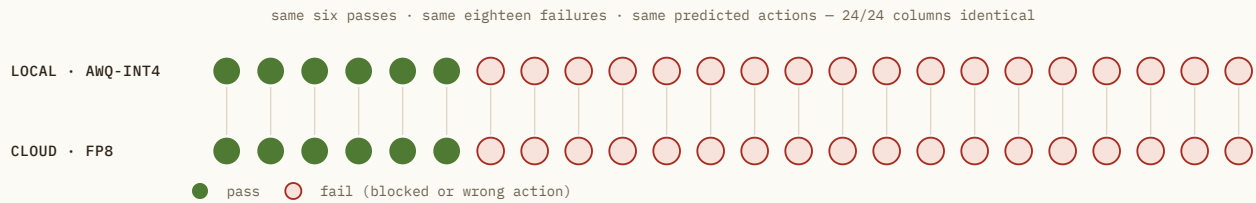
People react quickly. The process closes slowly.

THE PROOF

The benchmark was built to disappoint us

Claims are only as good as the instruments that check them, so the instruments came first: canaries gate every batch (one miss quarantines it), golden sets score known answers, and the replay benchmark asks the only honest question – given what was knowable that morning, can the system predict the action the human took? The blind slice has never been run.

First reveal: the local and hosted lanes matched on every canary, but only a quarter passed. That makes local results useful for screening, not promotion; what you fix locally still returns to the hosted lane for final validation.



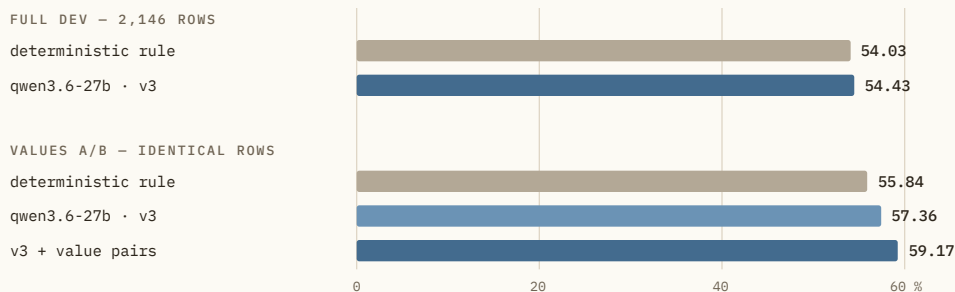
The quantization-equivalence check, drawn out. One column per canary case; columns sorted with passes first (which six passed is recorded in the canary report – the order here is for display). Six passes is a poor score; twenty-four identical columns is the point: a strong screening check that the cheap local lane tracks the hosted one, so local iteration is worth trusting before final hosted validation.

Second reveal: an early sample suggested an 82% result. Full scale brought it back to about 54%. The gates caught the mirage before anyone saw it as a claim.

► PROMPT LINEAGE, GENERATION BY GENERATION

The lab can run an experiment on its own; it cannot promote its own result. A lab loop runs a prompt generation end to end; a gate-check scores it against the promotion gates – pointed at the real full-dev results, it correctly returned *not passed*; a registry tracks pack lineage; a sentinel watches for drift. Promotion into the ledger remains a human-only action.

Third reveal: the model and the fixed rule both scored about 54% – a statistical tie. With richer safe evidence, the comparison moved to about 59% versus 56% – roughly a three-point early signal, and the next round's target.



Directional agreement with the historical human QA action, dev slice only. SE ≈ 2.5 points – every gap on this chart is within noise. No lane cleared the promotion gates, and the blind slice was never unlocked.

The project did not produce a model victory. It produced a measuring system willing to say so – the tie tells us the reasoning still needs improvement, and the roughly 3% of benchmark cases where model and rule disagreed became precedent candidates, not prompt fuel.

The benchmark measures both halves – read the two scorecards separately:

AI CAPABILITY	RESULT
Full benchmark	about 54% versus about 54% – a statistical tie
Richer safe evidence, identical rows	about 59% versus 56% – roughly a three-point early signal
Abstention on richer evidence	below 1% – the named target of the next revision
Per-class recall	a permanent gate – it is what killed an approve-machine

HARNESS RELIABILITY	RECORD
Parse success on the benchmark run	100%
Unsupported commitments escaped	zero
Data-safety violations	zero
Canary gate	one miss quarantines the batch – live on every run
Lane consistency	every local/hosted canary matched; 99% agreement across paired cloud runs
Blind test	sealed – never run

The model score tells us how capable the AI is. The harness record tells us whether that capability is safe to use.

A benchmark you cannot fail is marketing. This one can say "not yet." The blind slice stays locked until a lane earns it.

THE TAKEAWAYS

What this changed in how I build

Give AI the messy work; keep authority in the harness. The model navigates ambiguity. The harness decides what may survive.

Make evidence part of the interface. A recommendation is incomplete until its support travels with it.

Constrain structure; verify meaning. The schema shapes the answer. The verifier decides whether it is supported.

Turn settled judgment into infrastructure. Confirmed answers move repeat questions into code.

THE COMPOUNDING IDEA

The system gets smarter in two ways

Better evidence, prompts, and models improve the AI. Confirmed rulings improve the harness by moving repeat questions into code.

The AI gets better at unfamiliar cases. The harness makes familiar cases disappear.

The AI provides the intelligence. The harness turns it into a decision we can trust.

APPENDIX

Claim discipline: the numbers, sourced

House rule: state only what a file proves. Every figure in this post traces to the repository.

► THE PRODUCTION SYSTEM PROMPT, IN FULL

► BASELINE DETAIL: ISSUE MIX AND ACTIVITY PROXY

► EVAL APPARATUS: CANARIES, GOLDENS, REPLAY SPLIT

QUANTITY	VALUE	PROVABLE IN
QA-manual rulepack rules	25	rules/qa_manual_repair.v1.json
Handbook-seeded ledger entries	24	rules/equivalence_ledger.v1.json
Synthetic canary cases	24	examples/synthetic_cases.v1.jsonl
Golden cases, v1 / v2	10 / 51	examples/qa_repair_golden.v*.jsonl
Curated train / test rows	187 / 52	examples/datasets/
Authoritative catalog codes (result / comment)	20 / 532	WORKLOG.md · catalogs.py
Confidence floor for a committed decision	0.65	verifier.py
Replay split (dev / blind rows)	2,146 / 1,010	WORKLOG.md
Local int4 ↔ cloud fp8 per-case agreement	24/24 (6/24 pass)	WORKLOG.md · HANDOFF_GEN3
Paired cloud providers, raw agreement	94/95 · 98.9%	WORKLOG.md
Full-dev directional, rule / model v3	54.03% / 54.43% (SE ≈ 2.5)	WORKLOG.md
Model-rule disagreements → precedent candidates	65 of 2,146	WORKLOG.md
Test suite	343 tests	pytest --collect-only, 2026-07-09
QA ops baseline: window · finalized pool	2026-01-02→06-30 · 338,778	analysis/qa_ops_metrics_20260707/RESULT.md
Strict coded returns / clean approvals	6,453 / 195,974	RESULT.md · qa_ops_metrics_summary.json
Median return→rework / return→approve, trimmed	0.77 h / 192.8 h	qa_cycle_time_metrics.csv
Top-4 procedural share of coded return events	85.6%	qa_issue_category_counts.csv

Internal engineering note · 2026-07-03 · operational baseline added 2026-07-07. The pipeline described is `src/snh_ubs_qa_harness/`; the doctrine is `README.md` and `docs/DATA_SAFETY.md`.

The worked examples are synthetic. No applicant data, names, identifiers, or document references appear in this post – the same guarantee the harness enforces on itself, checked the same way.

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